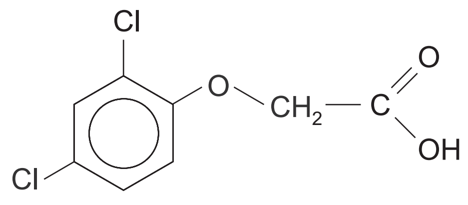
9. *(a)* A solid compound is believed to be the herbicide 2,4-dichlorophenoxyethanoic acid (2,4-D).



The following results were obtained when this compound was analysed.

- The percentage by mass of chlorine present was 32.1
- A solution of the compound reacted with sodium hydrogencarbonate to give a colourless gas
- 4.09 g of the compound was dissolved in a suitable solvent and the solution made up to 250 cm^3. A 25.0 cm^3 sample of this solution reacted with 24.70 cm^3 of aqueous sodium hydroxide of concentration 0.0750 mol dm^{-3}
- Refluxing a sample of the compound with aqueous sodium hydroxide did **not** show the presence of chloride ions in the resulting mixture
- The compound did **not** react with aqueous bromine to give a white precipitate or to decolourise the bromine
- The simplified ^1H NMR spectrum of the compound showed three separate peaks having the area ratio 1 : 2 : 3
- The ^{13}C NMR spectrum of the compound showed eight peaks

Examiner only

Use **all** the information to confirm that the compound is likely to be 2,4-D and suggest a simple laboratory method that would help to confirm that it is 2,4-D. [6 QER] *(b)* An ester can be hydrolysed by heating it with aqueous sodium hydroxide solution.

(i) Give the equation for the hydrolysis of methyl ethanoate. [1]

(ii) The hydrolysis reaction can be used in a quantitative way to find the relative molecular mass of an ester.

A known mass of an ester is refluxed with a known volume of sodium hydroxide solution of concentration 1.0 mol dm^{-3} and the excess sodium hydroxide remaining after hydrolysis is determined by an acid-base titration.

The following results were obtained.

Mass of ester used = 1.76 g

Volume of aqueous sodium hydroxide added = 50.0 cm^3 (V_2)

Volume of aqueous sodium hydroxide remaining after hydrolysis = 30.0 cm^3 (V_1)

I. Use the formula below to calculate the M_r of the ester. [1]

$$\text{mass of ester} = \frac{M_r \text{ of the ester} \times (V_2 - V_1)}{1000}$$

M_r = ...

II. The ester reacts with Tollens' reagent to give a silver mirror. Use this information and the M_r of the ester found in part I to suggest a formula for the ester. Give a reason for your answer. [2]

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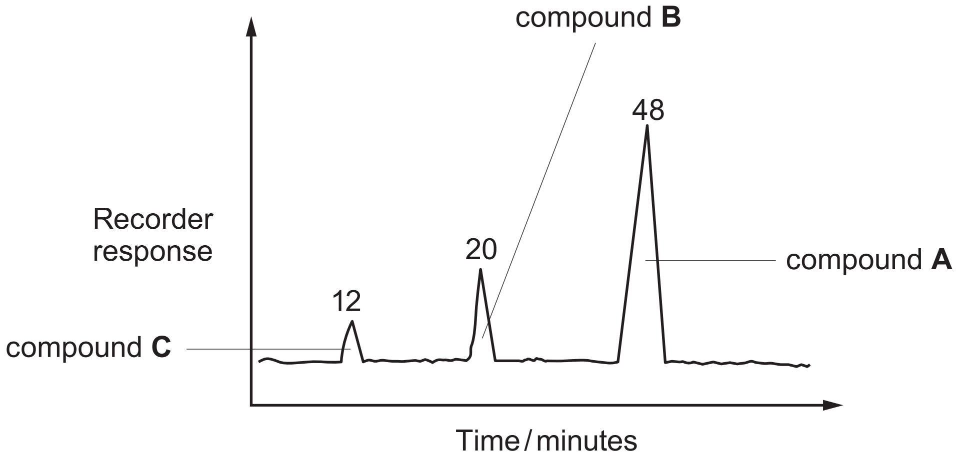
(iii) Esters of ethanoic acid with the formula $CH_3COOC_5H_{11}$ are used to give fruity flavours to confectionery.

I. The material used for this flavouring is often a mixture of esters having the formula above.

The diagram shows a gas chromatogram of a typical mixture.



Use the peak area data to calculate the percentage of compound **A** that is present in the mixture. [1]

Percentage = ... %

II. The peak areas shown give the molar percentage and in this case also the percentage by mass of each component present in the mixture.

Explain why the molar percentage of each compound **cannot** be used to find the percentage by mass of each component if the relative molecular mass of each component differs. [1]

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Examiner only

(iv) The boiling temperatures of compounds **A**, **B** and **C** in part (iii) are shown in the table.

Compound	Name	R in $CH_3-C(=O)-O-R$	Boiling temperature / °C
A	2-methylbutyl ethanoate	$-CH_2CH(CH_3)CH_2CH_3$	134
B	3-methylbutyl ethanoate	$-CH_2CH_2CH(CH_3)CH_3$	142
C	pentyl ethanoate	$-CH_2CH_2CH_2CH_2CH_3$	147

Suggest a reason for the difference in boiling temperatures between compounds **A** and **C**. [1]

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(v) Compound **A** is made by heating 2-methylbutan-1-ol and ethanoic acid in the presence of concentrated sulfuric acid.

State the name of another organic compound that could be formed if concentrated sulfuric acid reacted with 2-methylbutan-1-ol. [1]

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